

Modeling the alighting and boarding process through train doors using a Markov process with flow-dependent transition probabilities.

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Abstract

Understanding and modeling the alighting and boarding process in suburban train services is crucial to optimizing train dwellings. The alighting and boarding process is a bi-directional pedestrian flow through a bottleneck, being the door opening. Pedestrian flows, including alighting and boarding processes, are generally modeled by two-dimensional pedestrian models, such as cellular automata or social force models. These two-dimensional models are calibrated from two-dimensional data sources that are often complicated to access for privacy reasons. The availability of disaggregated passenger counting data led us to propose a different modeling approach based on cumulative flows. The model is a Markov process with variable transition probabilities. Transition probabilities are computed from the remaining number of alighting and boarding via a differential equation based on the pedestrian fundamental diagram and density estimations. The parameters of the differential equation were fitted using disaggregated passenger counting data. The model shows better predictive power than a linear benchmark model calibrated on the same data. The physical parameters of the model are consistent with the existing literature. The proposed approach offers an alternative to commonly used two-dimensional models, providing easier calibration. Such a model will enable the forecasting of alighting and boarding time distributions, facilitating better dwell time planning and train and platform design.

1. Introduction

In dense urban areas, railway transportation is increasingly plebiscite to move many people. People go from one station to another, making the train stations places of dense pedestrian flows. Pedestrians have various goals once arriving at a station (leaving the station, going to a platform to wait for a train, etc), and they move through multiple kinds of space (platforms, stairs, tunnels, halls, etc.). Although complicated, such pedestrian flows have to be correctly managed to guarantee pedestrian safety and comfort, as long as reliability for train operations. The alighting and boarding process is particularly critical and is unavoidable on any train journey. Therefore, alighting and boarding processes need to be accurately described, pedestrian models are necessary to do so.

Several approaches have already been proposed to model pedestrian flows. These approaches were grouped into nine main categories (Duives et al., 2013; Dong et al., 2019). These categories are: continuous models (Henderson, 1971; Hughes, 2002), cellular automata (Blue and Adler, 1998; Schadschneider et al., 2002), social force models (Helbing and Molnar, 1995; Xi et al., 2011), velocity-based models (Paris et al., 2007; Moussaïd et al., 2011), activity-choice models (Hoogendoorn and Bovy, 2004), macroscopic-microscopic hybrid approaches (Xiong et al., 2009), behavioral models (Robin et al., 2009; Wijermans, 2011), network models (Løvås, 1994; Daamen, 2002b) and AI approaches (Martinez-Gil et al., 2014; Song et al., 2018). From these works, simulators have been built dedicated to modeling pedestrian flows in various situations. Among them are: Nomad (Hoogendoorn and Bovy, 2004), an activity-based model that has been used, for example, to assess a railway station design (Hoogendoorn and Daamen, 2004), or SimPed (Daamen, 2002b) used to simulate pedestrians in stairs and tunnels (Daamen, 2002a), or platforms (Daamen and Hoogendoorn, 2004) in railway stations. More recent simulators include, among others, Simba mobi (Scherr et al., 2020), Trajectron++ (Salzmann et al., 2020), AgentFormer (Yuan et al., 2021), or MID (Gu et al., 2022). Commercial simulators have also been developed. Among them, one may mention VisWalk, MassMotion, or

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Legion. See, for example, Dubroca-Voisin et al. (2019) for a review. VisWalk is based on a social force model, while the other two used activity-choice approaches. The focus of this work is the alighting and boarding process.

From a pedestrian point of view, the alighting and boarding process consists of passing through a train door (consisting of a bottleneck with a limited width) in a bi-directional flow. For this more specific case, some of the above-mentioned approaches have been applied to simulate alighting and boarding flows. In general, social force models (Rudloff et al., 2011; Yu et al., 2019; Yang et al., 2020; Su et al., 2020; Yang et al., 2021; Li et al., 2020) or cellular automata (Zhang et al., 2008; Sun et al., 2021; Qu et al., 2023) are chosen to model the alighting and boarding process. For example, Li et al. (2020) derived a social force model with particular attention to the influence of people boarding before all alighting people had alighted (known as non-compliance behavior). Zhang et al. (2008) used a cellular automata model to consider individual differences to model the process. The alighting and boarding process is limited in time and space, which makes the dynamics insensitive to strategic and tactical pedestrian choices. Activity-choice models, velocity-based models, or behavioral models are then less suitable for alighting and boarding simulation.

In addition, like any human behavior phenomenon, the alighting and boarding process is subject to important variability, which is important to be modeled to manage dwell time and prevent dwell time delays. Buchmueller et al. (2008), for example, measured the different sub-processes of the dwell time from Swiss field data and showed that most of the variability in dwell times came from the alighting and boarding processes (even with a given number of movements). Similar results were obtained by Seriani et al. (2019) in a laboratory experiment. Both saw non-trivial variability patterns: Buchmueller et al. (2008) observed that alighting and boarding time variability increases more slowly than the number of passenger movements. Seriani et al. (2019) obtained decreasing per-passenger variability with time. However, none of these references commented much on these variability patterns. Every individual is different from one another (inter-individual variability), and people's behaviors may change according to the context (intra-individual variability). This variability is inherent in the pedestrian flows involved. Therefore, microscopic approaches such as social force models or cellular automata were generally researchers' choices to reproduce the variability of the flow. However, network-based models are also able to consider stochasticity in the process.

However, pedestrian approaches to model the alighting and boarding process were rarely calibrated with precise data. Some models were not calibrated from data at all, using parameters from the literature to study specific influences of certain features (Li et al., 2020; Yang et al., 2021; Sun et al., 2021). Some models were calibrated on aggregated data, generally alighting and boarding times (Zhang et al., 2008; Yang et al., 2020; Su et al., 2020). When calibrated with precise data, the most common choice was video data (Rudloff et al., 2011; Yu et al., 2019; Qu et al., 2023), as discussed in the review by Haghani and Sarvi (2018).

At the same time, it is sometimes difficult to access and use video cameras for privacy reasons. Thus, we propose a novel approach to model the alighting and boarding process based on passenger counting data that are increasingly available in large quantities in some train lines. Several works have already used this kind of data for modeling purposes, but, to our knowledge, only to model minimum dwell times, or alighting and boarding times (Cornet et al., 2019; Coulaud et al., 2023). Contrary to existing approaches, which are based on two-dimensional representations of pedestrians around the train door, we propose to model the alighting and boarding process from cumulative flows, already discussed for pedestrians in Daamen et al. (2005), for example. Our goal is to use counting data at the most disaggregated scale possible (here at the door counting event scale, number of people that have passed every several seconds), to calibrate a cumulative flow model of the alighting and boarding process. This level of resolution for real-world data has rarely been used so far, with an exception in the study of bus alighting and boarding times using automatic fare counting data (Sun et al., 2014). Disaggregated counting data provide information on the time each person crosses the door, but no information on the spatial distribution of the passengers on both sides of the door. This information is, yet, crucial for the existing approaches. Describing the alighting and boarding process from this piece of data then requires an innovative approach.

To our knowledge, only one kind of model has been proposed so far in the literature to model pedestrian cumulative flows: queuing models. This is, for example, the case in the work of Løvås (1994) to get from one room to another in a network-based model. However, Løvås (1994) assumes constant queue rates (independent of any flow or density condition). This kind of model will serve as a benchmark in our study because it can be fitted on our real-world data, but our cumulative flow model will also consider density conditions.

The flow-density relationship for pedestrians is described by so-called fundamental diagrams. The oldest example for pedestrians is probably Hankin and Wright (1958), but the concept had been used for a long time for cars (Greenshields et al., 1935). Many relationships have been proposed since (Weidmann, 1993; Hurley et al., 2015).

Vanumu et al. (2017) proposed a review of existing fundamental diagrams in various contexts (uni-directional and bi-directional flows, bottlenecks, stairs, etc) and discussed the resulting differences. For alighting and boarding processes, the study of Daly et al. (1991) is an example of a fundamental diagram plot from experimental data.

In this work, we will use disaggregated counting data to fit varying alighting and boarding rates depending on estimated density conditions. The fitted values will serve as an average rate for a stochastic queuing model to represent the cumulative flow of alighting and boarding. This work aims to model pedestrian flows at train doors with an original stochastic approach: a queuing model with varying alighting and boarding rates depending on the density conditions. The model is fitted on disaggregated counting data.

In summary, our contributions are:

- Proposing a cumulative flow deterministic model calibrated on disaggregated passenger counting data to model the alighting and boarding process,
- Proposing a stochastic cumulative flow model that is a generalization of the deterministic model,
- Estimating the alighting and boarding times distribution using the stochastic model.

First, we precisely present the data source, the data selection, and the delimitation of the perimeter. Next, the method is presented, going from the general idea to the precise formulas used here. The following section is dedicated to the results showing the good ability of the model to describe the alighting and boarding process and the alighting and boarding time compared to a linear benchmark from a 5-fold cross-validation computation. The sensitivity to the data is also discussed, and the physical meanings of the parameters are interpreted. This article ends with a conclusion.

2. Data selection and perimeter

In this work, we propose a cumulative flow model for the alighting and boarding process. One of the major innovations of our approach is the use of disaggregated passenger counting data to calibrate and validate the model. This data is first introduced before describing the model in further detail in Section 3.

Disaggregated passenger counting data is provided by infrared sensors above each train door that count the cumulative numbers of alightings and boardings. This technology is available for a new rolling stock called Regio2N. Every few seconds, information on the cumulative number of alightings $A(t)$ along with the cumulative number of boardings $B(t)$ is registered. We name this information a counting event. The time separating two counting events is truly random with an average of 3s and a standard deviation of 3s. The system also registers for each door whether the door is open or closed. A study from the data provider assessed the quality of the sensors using manual counting. They found a root mean square deviation of 4.4 people boarding and 3.6 people alighting. Such deviation is quite high, but mainly corresponds to variance, having a bias of +0.13P in alighting, and +0.24P in boarding. The amount of data should then mitigate the effect of data quality in the fitting. We will confirm this statement in Section 4. The dataset is over six months (September 2022 to February 2023), where we have data for almost 30,000 stops (see Table 2).

Each unit of the Regio2N (see Figure 2) is an articulated train composed of eight cars. Not all cars are fully similar:

- cars 3, 5, and 7 are identical, contain two doors, and are single-deck;
- cars 2, 4, and 6 are identical, do not contain doors, but are double-deck;
- cars 1 and 8 contain one door and are not identical.

Furthermore, except for cars 1 and 8, the cars are symmetric. In addition, the eight doors of the train are 1.60 meters wide, and the doors of cars 1 and 8 (that is, door 1 and door 8) are equipped with sliding platforms. Existing works on alighting and boarding times showed a dependency on the train design. Harris (1994) considered the width of the door and the number of seats in the carriage. Su et al. (2020) showed an influence on the proportion of seats inside the train based on simulation. The influence of door width is regularly highlighted in the literature on bottlenecks (Seyfried et al., 2009), and an example for buses is also provided by Fernández et al. (2010). More generally, the rolling stock design is considered a variable influencing the alighting and boarding process (Thoreau et al., 2016; Christoforou et al., 2020). Thus, we only considered doors 3 to 6, where the interior designs related to them are symmetrically similar. Moreover, Daamen et al. (2008) showed an influence of vertical and horizontal gaps between the train and the platform. Doors 3 to 6 are identical in this respect and without sliding platforms.

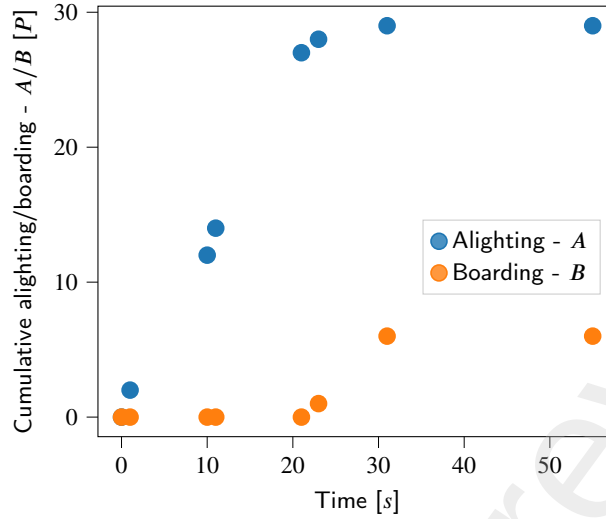


Figure 1: Example of cumulative data provided by the infrared sensors for one stop and one door. Time 0 corresponds to the end of the opening of the door.

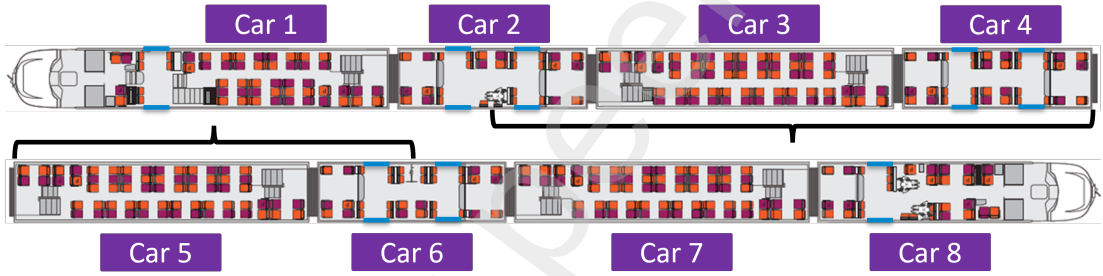


Figure 2: Design of the Regio2N rolling stock. Parts marked with braces are selected.

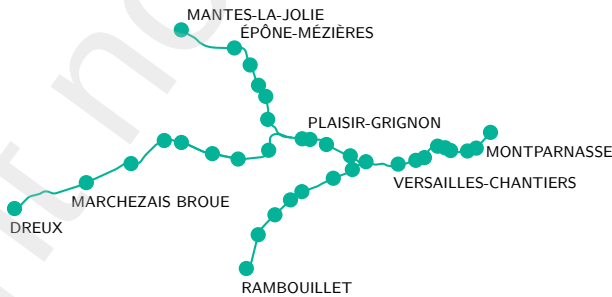


Figure 3: Map of line N. Commuter train line in the Paris (France) suburban area.

Line N of the Paris suburban network is operated only by Regio2N. Figure 3 shows the map of this line, from which the data for this study come. The platforms in this line are the same height as the train floor, except for platforms at *Marchezais Broué* and *Epône-Mézière* where it is lower than the other stations. Both stations were excluded due to the influence of vertical gaps on the alighting and boarding process (Daamen et al., 2008). The influence of vertical gaps or steps was also proven for buses (Fernández et al., 2010; Tirachini, 2013).

Line N is a commuter line: it is unlikely to see people with lots of luggage, whose influence (shown by Daamen et al. (2008) or Kuipers et al. (2025)) can, thus, be neglected. The volume of passengers in this line is intermediate, with an

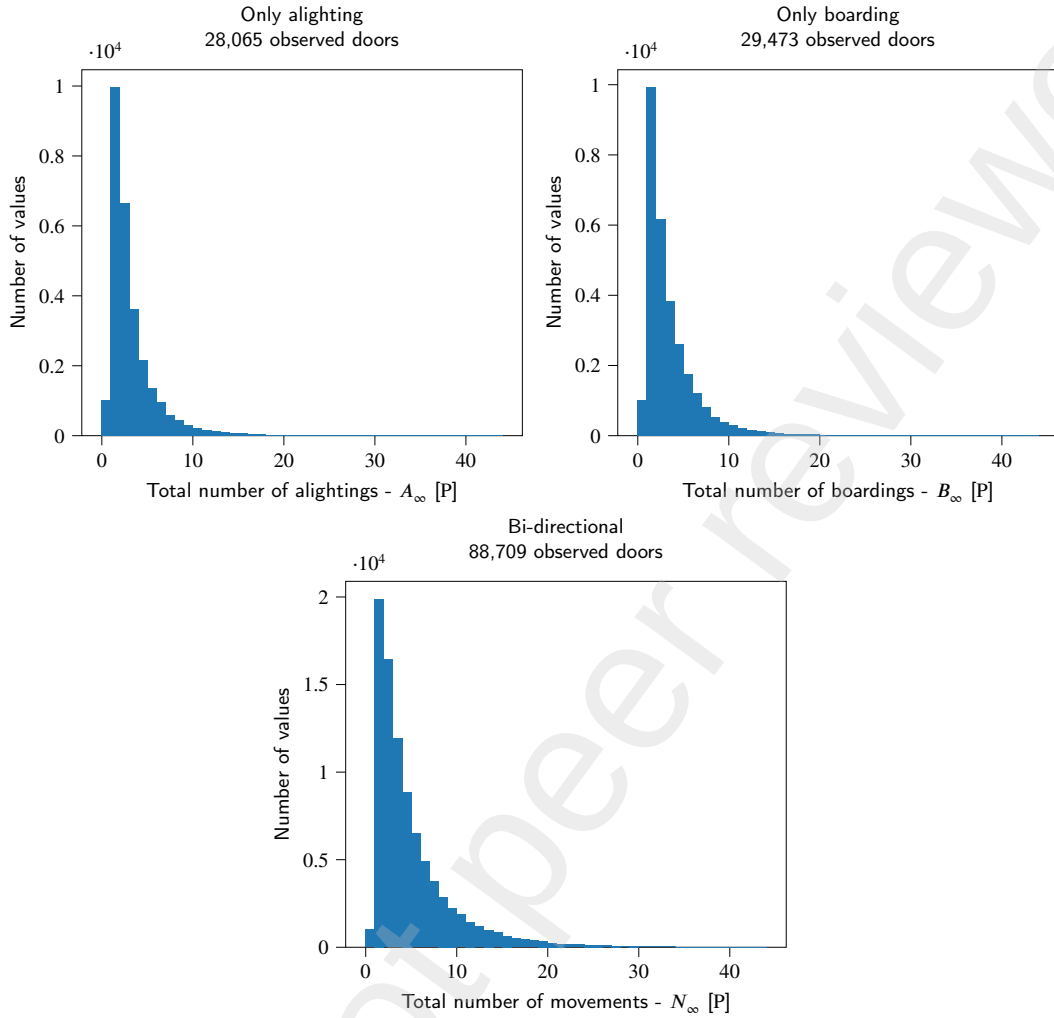


Figure 4: Histograms of total numbers of alightings (top left), boardings (top right), and movements (bottom) for uni-directional flows (top) and bi-directional flows (bottom) for any open door. Most alighting and boarding processes are composed of only a few movements.

average load of 169 passengers, much less than the capacity (1054 passengers). Furthermore, the seating capacity (524 passengers) is exceeded 0.2% of the time. Then, we can neglect the impact of on-board passengers (neither alighting nor boarding), though proven qualitatively important for alighting and boarding modeling considering congestion levels (Seriani and Fernandez, 2015; Seriani et al., 2019), and quantitatively considering their number (Harris, 1994; Cornet et al., 2019).

In this perimeter, we consider three situations: only alighting, only boarding, and bi-directional flows. The total number of alighting and boarding at each door is a widespread variable in modeling alighting and boarding times (Harris, 1994; Cornet et al., 2019; Coulaud et al., 2023) and alighting and boarding rates (Harris et al., 2022; Kuipers et al., 2025). The relationship with the total number of alighting and boarding often includes non-linearities (Harris, 1994; Cornet et al., 2019; Harris et al., 2022; Kuipers et al., 2025) but also a dependency on the boarding over alighting ratio (Seriani et al., 2019; Fu et al., 2023). For example, in non-crowded situations Fu et al. (2023) showed significantly higher alighting and boarding rates with ratios of 0.25 or 4 compared to a ratio of 1 when passengers complied with the alighting priority. In contrast, the opposite result was obtained when passengers did not comply with the alighting priority. For a first approximation, we will consider any passenger movement (alighting or boarding) without distinction in the bi-directional case and neglect the boarding/alighting ratio.

In each situation, the number of movements for a given door is rarely higher than 20 movements; see Figure 4. With such volumes, no congestion is likely to happen in the train or on the platform. In particular, the alighting area on the platform does not influence the alighting and boarding process, as shown by Yang et al. (2020) from simulations.

Furthermore, we consider only the alighting and boarding process with more than six alighting people or boarding people in the case of uni-directional flows, more than six movements (alighting plus boarding) with at least one alighting and one boarding in the case of bi-directional flows. On the one hand, having too few passengers involved in the process may give too much weight to their inter-individual differences and variability in their behavior, let alone sensors' errors. On the other hand, we need to have enough events in the dataset for the study. The number six is a trade-off for a decent consideration of both restrictions.

Passengers have several kinds of inter-individual differences that may impact the alighting and boarding process modeling. For example, passenger ages (Tirachini, 2013), genders (Zhang et al., 2008), or body shapes (Yang et al., 2021) were shown to influence the process. Other aspects related to variability in passengers' behaviors were studied in the literature, such as passengers' ability to compete to get in or out of the train (Zhang et al., 2008) or their individual desired speed (Yang et al., 2021). All the variability due to passengers' differences or behaviors will be considered as random in this work. More details on how we model randomness will be provided in Section 3.

Among the remaining data, other subsets will be considered:

- lower non-compliance and higher non-compliance to alighting priority, shown to have a weak but non-negligible effect (Wahaballa et al., 2022; Seriani et al., 2022) (we consider higher non-compliance when at least 3 boarding passengers board before 90% of alighting people have alighted, lower non-compliance for 2 or fewer non-compliant boardings);
- peak hours (6 am to 10 am, 4 pm to 8 pm) and off-peak hours, where it is likely to see different patterns occurring, but not always observed (Christoforou et al., 2020);
- the specific station of *Versailles-Chantiers*, main station in *line N* except for the Paris terminus, where differences may occur due to station differences or higher demand on average (Qu et al., 2019; Christoforou et al., 2020).

Several determinants of the alighting and boarding process discussed in the literature are constant in our perimeter or in specific subsets of the data. Table 1 summarizes them. Other factors will be considered as variables in the model or as random.

Table 2 summarizes the different steps of the data selection and the studied subsets of data, along with the remaining data quantities, average number of movements, and average flows. Only a little variation is observed in the average numbers of movements and flows between the studied subsets.

3. Method

As already mentioned, this work aims to propose a model describing the alighting and boarding process from a cumulative flow approach. Formally, we aim to get the dynamics of the cumulative alighting $A(t)$ versus time, the cumulative boarding $B(t)$ versus time, and the cumulative passenger movements $N(t)$ versus time at the door level. First, we derive a differential equation to get a deterministic model, which is the average dynamic of the stochastic model. Then, we include randomness, considering the alighting and boarding process as a Markov process (with varying rates depending on the estimated density condition). Both the deterministic part and the stochastic part will be assessed separately.

3.1. Deterministic modeling of the alighting and boarding process

First, we derive deterministic differential equations to obtain the average dynamics of the alighting and boarding process. Our approach differs from macroscopic two-dimensional approaches in several ways, which we will detail in this section. Table 3 summarizes them.

3.1.1. General differential equation system

For the sake of simplicity, we will note X any of A , B , or N . For better readability, we will not write all the indices concerning the stop and the door (train number, station, date, door number), and only write $X(t)$. $t = 0$ is set at the end of the door opening. The boundary conditions are then $X(t = 0) = 0$ and $X(t = +\infty) = X_\infty$, where X_∞ is the total number of alighting people, or boarding people, or movements at this door and this stop.

Feature	Reference	This approach				
		Variable	Constant	Neglected	Random	Sub-sampled
Demand						
Alighting	Harris (1994); Coulaud et al. (2023)	X				
Boarding	Harris (1994); Coulaud et al. (2023)	X				
On-board	Harris (1994); Cornet et al. (2019)			X		
Flow features						
On-board congestion	Seriani and Fernandez (2015); Seriani et al. (2019)			X		
Non-linearities	Cornet et al. (2019); Kuipers et al. (2025)	X				
Boarding/alighting ratio	Seriani et al. (2019); Fu et al. (2023)				X	
Rolling stock features						
Number of seats	Harris (1994); Su et al. (2020)		X			
Effective door width	Harris (1994); Fernández et al. (2010)		X			
Rolling stock type	Thoreau et al. (2016); Christoforou et al. (2020)		X			
Platform features						
Horizontal gap	Daamen et al. (2008)		X			
Vertical gap	Daamen et al. (2008); Fernández et al. (2010)		X			
Alighting area width	Yang et al. (2020)			X		
Station	Qu et al. (2019); Christoforou et al. (2020)					X
Passenger-related features						
Peak/off-peak hour	Christoforou et al. (2020)					X
Non-compliance	Wahaballa et al. (2022); Seriani et al. (2022)					X
Luggage	Daamen et al. (2008); Kuipers et al. (2025)			X		
Ages	Tirachini (2013)				X	
Genders	Zhang et al. (2008)				X	
Body shapes	Yang et al. (2021)				X	
Competition	Zhang et al. (2008)				X	
Desired speed	Yang et al. (2021)				X	

Table 1

Features determining the alighting and boarding process considered in the literature and their treatment in our approach.

Selection step or subset	Number of stops	Number of counting events	Average number of movements [P]	Average flow [P · s ⁻¹]
All counting data on <i>line N</i>	29,768	764,263	4.87	0.41
Only doors 3 to 6	27,839	379,725	4.59	0.40
All stations but <i>Marchezais Broué</i> and <i>Epône-Mézière</i>	27,357	375,465	4.61	0.40
More than six people alighting (unidirectional case)	2,347	14,892	9.10	0.52
More than six people boarding (unidirectional case)	2,634	19,752	8.87	0.40
More than six people of both (bidirectional case)	8,347	98,841	10.20	0.58
Peak hours	4,347	53,744	11.71	0.60
Off-peak hours	4,000	45,097	10.60	0.57
Lower non-compliance	7,955	85,127	10.88	0.59
Higher non-compliance	1,606	13,714	13.70	0.56
Only <i>Versailles-Chantiers</i>	1,517	27,928	13.77	0.63

Table 2

Summary of the steps in the data selection and the studied subsets, along with the remaining data quantities and the average number of movements and flow. Only a little variation is observed in the average numbers of movements and flows between the studied subsets. Notice that a stop is not considered if all remaining doors have 0 alightings and 0 boardings.

A traditional way to analytically describe pedestrian flows is to use a fundamental diagram and a conservation law. The main assumption behind our approach, a cumulative flow approach, is that the speed v only depends on the density ρ ($v = v(\rho)$), ρ will then be estimated from another equation. From this, in a time-dependent situation, we have $\phi(t) = \rho(t)v(\rho(t))$. A conservation law is not relevant in our case since it is not a two-dimensional problem, but only a cumulative flow model. Instead, we derive a differential equation of X to adapt this relation to the cumulative flow modeling.

	This approach	Macroscopic 2D approach
Use case	Crossing (here train doors)	General
Data	High-resolution counting	Video
Variables	Cumulative flows	Flows & Densities
Access to density	Implicit	Explicit
Equation associated with fundamental diagram	Density function	Conservation law

Table 3

Differences between our approach and a two-dimensional macroscopic approach.

We have, from the definition of cumulative flows: $D\phi(t) = \frac{dX}{dt}$ (where D is the door width) which means: $\frac{dX}{dt} = D\rho(t)v(\rho(t))$. Densities should vary with time. In particular, as people alight/board, the pressure inside/outside the train should decrease, leading to a decrease in density. Thus, we postulate that $\rho(t)$ only depends on $X(t)$: $\rho(t) = f(X_\infty - X(t))$. f is the density function that serves to estimate ρ from X .

This leads to an ordinary differential equation:

$$\frac{dX}{dt} = Df(X_\infty - X(t))v(f(X_\infty - X(t))). \quad (1)$$

Table 4 summarizes the important variables in this work.

One may object that the alighting and boarding process is a bi-directional flow. Yet, most of the time, fundamental diagrams for bi-directional flows consider the total flow (sum of the flow from both directions) versus the total density (density from both directions) as shown in the review of Vanumu et al. (2017). In the alighting and boarding process, a common rule is to give priority to alighting. Some boarding passengers may not comply with this rule, and such behavior cannot be modelled with flows and densities. However, Baali et al. shows that compliance with this rule has little impact on cumulative movement dynamics ($N = A + B$) with low to intermediate total movements and a balanced boarding/alighting ratio overall. This counterintuitive result is due to similar flow between a density below the optimal density (density resulting in the maximal flow) with lower compliance and a density higher than the optimal density with higher non-compliance. The approach proposed in this work does not enable the prediction of whether and when non-compliance behavior may occur and, consequently, the dynamic of the boarding/alighting ratio versus time. Then, we will neglect the effect of both non-compliance and boarding/alighting ratio and keep the same form of ordinary differential equation for N .

In the next subsection, the function v will be chosen among the fundamental relations proposed in the literature, and f will be conjectured from the previous assumptions. v and f are assumed to be constant throughout the dataset. The differential equation obtained will be solved to obtain A , B , or N versus t .

3.1.2. Choices for evolution equations

As presented before, two relations have to be set to complete the differential equation: the fundamental diagram and the density function. This work aims to assess the method presented above, using one example for both of these relations. The choice of these relations is qualitatively discussed, and future works should explore other forms of relations.

Fundamental diagram We select the Greenshields linear relation for the fundamental diagram, as often proposed in the pedestrian literature (Fruin, 1971; Pauls, 1980). Its expression is the following:

$$v(\rho) = v_0 \left(1 - \frac{\rho}{\rho_{max}} \right), \quad (2)$$

where v_0 is the free-flow speed, ρ_{max} is the maximal density. An illustration of this relation is presented in the left part of Figure 5. Other formulations have been proposed in the pedestrian literature, such as Underwood formulations (Lam et al., 1995; Rastogi et al., 2013), piecewise formulations (Polus et al., 1983), or non-linear formulations with additional parameters (Weidmann, 1993). The Greenshields formulation is a linear relationship with two physically interpretable parameters. This simple formulation was shown to be unsuitable for high densities (Helbing et al., 2007). From the volumes observed in the perimeter studied, it is unlikely that high densities occur. Then, a Greenshields

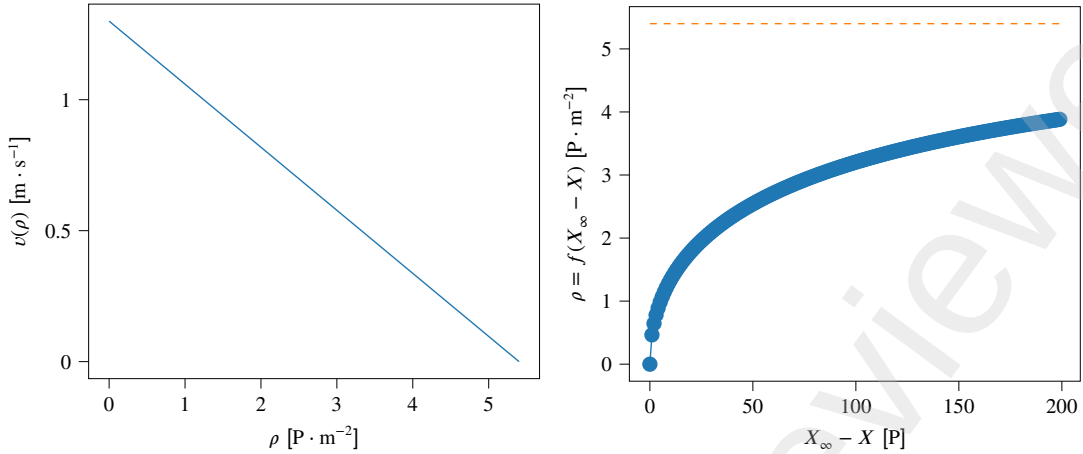


Figure 5: Illustration of the chosen fundamental diagram (left), illustration of the chosen density function (right), for $X_\infty = 200\text{P}$.

formulation would not lead to significant inaccuracies, and no data with sufficient volume would be able to fit high densities. Therefore, we focus only on the Greenshields formulation.

Density function Concerning the density function f , we set four assumptions.

1. The more people there are left to alight/board, the higher will be the density. Then f must be increasing.
2. A maximal density ρ_{max} cannot be exceeded, and the space in the train/the platform is limited. Then, f should tend to this maximal density, $\lim_{X_\infty - X \rightarrow \infty} f(X_\infty - X) = \rho_{max}$.
3. For no people, the density is zero: $f(0) = 0$.
4. When there are few people, people tend to accumulate around the door, increasing the density faster. Then, we assume f locally strictly concave close to 0: $\frac{d^2 f}{d(X_\infty - X)^2} > 0$. This assumption is inspired by the observations made by Wu and Ma (2013) concerning the positioning of boarding people on the platform.

Given those assumptions, we propose the following form for this relation:

$$f(X_\infty - X(t)) = \rho_{max} \left(1 - \exp(-\gamma \sqrt{X_\infty - X(t)}) \right). \quad (3)$$

An illustration is presented on the right part of Figure 5.

Obtained differential equation The obtained differential equation is therefore:

$$\frac{dX}{dt} = Dv_{0,X} \rho_{max,X} \left(1 - \exp(-\gamma_X \sqrt{X_\infty - X(t)}) \right) \left(\exp(-\gamma_X \sqrt{X_\infty - X(t)}) \right), \quad (4)$$

see Table 4 for a reminder of variables and parameter definitions.

This equation has only two degrees of freedom ($\psi_X = Dv_{0,X} \rho_{max,X}$ and γ_X), it is then more appropriate to write:

$$\frac{dX}{dt} = \psi_X \left(1 - \exp(-\gamma_X \sqrt{X_\infty - X(t)}) \right) \left(\exp(-\gamma_X \sqrt{X_\infty - X(t)}) \right). \quad (5)$$

3.1.3. Parameters fitting

From the equation presented above, we will get the best parameters from a fit. A traditional quadratic cost function would not lead to fit the best parameters. Errors when many people remain to alight propagate in the cumulative flow. In addition, processes with low volumes of passengers are the majority (see Figure 4). Then, we added a factor to penalize the cases at low remaining volume. Therefore, the cost function is a traditional quadratic cost function with the addition of a factor increasing the weight of the term proportional to the number of people remaining to alight (or board, or both).

Name/Expression	Symbol	Unit	Type
Cumulative alighting	A	[P]	Variable
Cumulative boarding	B	[P]	Variable
Cumulative sum of alighting and boarding	N	[P]	Variable
Total alighting	A_∞	[P]	Variable
Total boarding	B_∞	[P]	Variable
Total sum alighting and boarding	N_∞	[P]	Variable
Replace A , B , or N	X	[P]	Variable
Free flow speed	v_0	$[m \cdot s^{-1}]$	Hidden parameter
Maximal density	ρ_{max}	$[P \cdot m^{-2}]$	Hidden parameter
Door width (1.6m)	D	[m]	Fixed parameter
Density calibration parameter	γ	$[P^{\frac{1}{2}}]$	Fitted parameter
$v_0 D \rho_{max}$	ψ	$[P \cdot s^{-1}]$	Fitted parameter

Table 4

Summary of variables and their units.

Then, we minimize the cost function C_X :

$$C_X = \sum_{n=(k,s,d,j,t)} \left(\left(\frac{dX}{dt} \right)_n^{real} - \psi_X \left(1 - \exp(-\gamma_X \sqrt{X_{\infty,n}^{real} - X_n^{real}}) \right) \left(\exp(-\gamma_X \sqrt{X_{\infty,n}^{real} - X_n^{real}}) \right) \right)^2 \times (X_{\infty,n}^{real} - X_n^{real}) \quad (6)$$

The same cost function is considered to fit parameters for A , B , and N . The minimum is obtained using the function *minimize* from *scipy.optimize* in Python. The fit is sensitive to initial conditions. Then, we did 500 fits where we randomly chose initial conditions and selected the best solution overall. The data used corresponds to different subsets presented in Section 2. These data subsets are from similar enough setups to fit a single set of parameters for each subset. The parameters then feed the stochastic model.

3.2. Stochastic modeling of the alighting and boarding process

3.2.1. Alighting and boarding process: a Markov process

The deterministic model presented in the previous section aims to describe the average dynamics of the alighting and boarding process. Any passenger movement is yet subject to important variability due to various causes. A stochastic approach is presented in this section, to model the variability of time between two alightings, two boardings, or two movements: variability due to individuals (body shape, desired speed...), variability due to interaction between individuals (distance between individuals), variability due to irregular organization (variability of the density). All these variabilities are considered together in a stochastic process: $(X(t))_{t \geq 0}$.

$(X(t))_{t \geq 0}$ is assumed to be a continuous-time Markov process: moving from state $X(t) = i$ to another state only depends on the flow conditions at state $X(t) = i$. The only accessible state from state i is $i + 1$, except for X_∞ , which is absorbing. $X(t)$ can be seen as a birth process:

$$P(X(t+h) = j | X(t) = i) = \delta_{i,j} + \delta_{i+1,j} q_{i+1} h + o(h), \quad (7)$$

where δ is the Kronecker symbol.

Then, the time separating the i^{th} alighting and the $(i+1)^{th}$ alighting follows an exponential distribution whose mean $(\frac{1}{q_{i+1}})$ is given by the above-derived model:

$$q_i = \psi_X \left(1 - \exp(\gamma_X \sqrt{X_\infty - i}) \right) \left(\exp(\gamma_X \sqrt{X_\infty - i}) \right), \quad (8)$$

where ψ_X and γ_X are the parameters fitted for the deterministic model. It is not a Poisson process because the transition rate q_i is not constant with i a priori. An important observation to be made here is that the above-proposed deterministic model is the average of the proposed stochastic model.

This stochastic model can be seen as a queueing model where arrivals do not follow a Poisson process but a Markov process with varying transition probabilities (due to varying densities), and the service time is infinite.

Such a model enables the simulation of various alighting and boarding processes at the scale of individuals. In practice, this model is particularly interesting to evaluate the alighting and boarding time distribution.

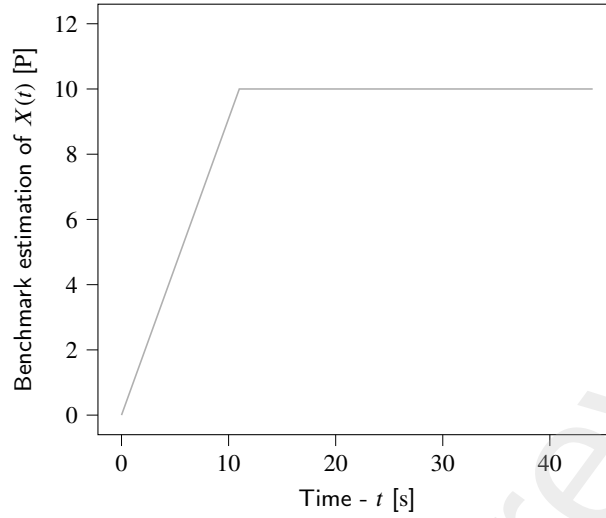


Figure 6: Illustration of the linear model selected as the benchmark in this work for A , B or N , in the case of $X_\infty = 10P$.

3.2.2. Simulating alighting and boarding time distributions

The proposed stochastic model is used to simulate the alighting and boarding processes from which we compute the alighting and boarding times. The method used to compute the alighting and boarding time is the quantile method developed by Baali et al. (2025) using the parameter $q = 0.8$.

From a given number of alighting (A_∞), boarding (B_∞), or total (N_∞), 1,000 alighting and boarding processes are simulated. The resulting distribution of alighting and boarding times is then compared to the distribution obtained from the real-world data (data from the sensors).

3.3. Benchmark model

In the literature, alighting and boarding times are often assumed to be linear with the number of alighting and boarding for practical use (D'Acerno et al., 2017; Medeossi and Nash, 2020). This linear model is often taken as a reference for comparison with other models, in particular for buses (Aashtiani and Iravani, 2002; Sun et al., 2014). In this study, we also consider a linear model as a benchmark. A determinist benchmark is as follows:

$$X = \begin{cases} \eta t, & \text{if } \eta t \leq X_\infty \\ X_\infty, & \text{otherwise} \end{cases} \quad (9)$$

where η is fitted the same way as the deterministic model. An illustration is proposed in Figure 6.

A stochastic benchmark is also considered, which is a queueing model M/D/1 with an infinite service time. Moving from state $X(t) = i$ to another state only depends on state $X(t) = i$, and the only accessible state from state i is $i + 1$, except for X_∞ , which is an absorbing state.

$$P(X(t+h) = j | X(t) = i) = \delta_{i,j} + \delta_{i+1,j} \eta h + o(h). \quad (10)$$

where η (the service rate) is the parameter of the deterministic benchmark model. This model is similar to the one proposed by Løvås (1994). The same observation can be made here: the deterministic benchmark is the average of the stochastic benchmark.

The stochastic benchmark is also simulated 1,000 times to obtain the alighting and boarding time distribution for fixed A_∞ (or B_∞ or N_∞).

Notice that two-dimensional models, such as cellular automata or social force models, would have been relevant benchmarks. However, given the absence of two-dimensional data sources (for example, videos) to calibrate them, we have not selected them.

3.4. Model assessment

To assess the model and the benchmark, we shuffled the different stops in the data, divided it into five samples, and performed a 5-fold cross-validation (with 4/5 of the data being the train set and 1/5 of the data being the test set).

The quality of the fit is assessed on $X(t)$ over the whole test data sample. The statistical indicators used are the mean absolute error (MAE), the root mean square error (RMSE), and the cost function.

A MAE of $1 \text{ P} \cdot \text{s}^{-1}$ means that, on average among the counting events, the model is $1 \text{ P} \cdot \text{s}^{-1}$ away from the real alighting rate, or boarding rate, or movement rate. The RMSE gives more importance to the counting events where the model is further away from the real data. Values of MAE and RMSE, along with the cost functions, are averaged over the five test samples of the 5-fold cross-validation. A confidence interval at 5% threshold is also computed to assess the significance of the difference between the model and the benchmark fitting.

Both the model and benchmark are compared with the alighting and boarding time distributions from the real data. The Kullback-Leibler divergence (Kullback and Leibler, 1951) between the model (or the benchmark) and real data is used for the comparison. Another assessment is done based on the 2nd and 8th deciles of the distributions because the main interest for practical use is the central part of the distributions, let alone the existence of some outliers in the real-world data.

Concerning the parameters obtained from the fit, their averages over the 5-fold cross-validation are compared to values obtained in the literature. The standard deviation among the five values is used to calculate confidence intervals at a 5% threshold.

If the deterministic model is assessed via a 5-fold cross-validation, it is not the case with the stochastic model. This is because too small amounts of data are available to build distributions for each total number of movements. This leads to correlations between our model's distributions and distributions from real data.

4. Results

4.1. Visual assessment

4.1.1. Visuals of the deterministic models

Figure 7 shows an example of the profile of cumulative alighting, cumulative boarding for uni-directional flows and cumulative movements for bi-directional flows obtained with the deterministic model (once the differential equation is solved), the deterministic benchmark, and the real data for the different occurrences having the same total number of alighting, boarding, or movements. With a higher volume (see right of the figure), despite important spread of the real data, the model seems to fit the alighting and boarding process better than the benchmark. With a smaller volume (left of the figure), little difference is observed between the benchmark and the model: both correctly fit the real data. In any case, the benchmark seems to better fit the beginning of the alighting and boarding process. This is probably due to the weighting term in the cost function of the fit, which provides more importance to cases where many people are left to alight and/or board (i.e., the beginning).

4.1.2. Visual of the stochastic model

A similar qualitative analysis can be done for the stochastic model and the stochastic benchmark. For a given cumulative flow X , the 2nd and the 8th deciles of t are plotted for real data, the stochastic model, and the stochastic benchmark (see Figure 8). The figure only shows the bi-directional case because insufficient data were available for uni-directional cases to get robust deciles. Both the stochastic model and the stochastic benchmark correctly reproduce the spread of the alighting and boarding process. The model shows superiority for a higher total number of movements (left of the figure). The stochastic model fits particularly well the end of the alighting and boarding process in both cases, where the spread increase is more important. In contrast, the stochastic benchmark seems to better fit the beginning of the alighting and boarding process, similar to what was observed in the deterministic case.

4.2. Quality of the fits

Then, we assess the performance of the proposed deterministic model compared to the deterministic benchmark. Table 5 shows these performances with MAE, RMSE, and the cost function, along with the relative confidence intervals. It is not worth assessing the stochastic model and the stochastic benchmark with these averaged indicators because the average of the stochastic model and benchmark corresponds to their deterministic counterparts. The results in this section mainly provide insight into the quality of the fit.

For the uni-directional alighting and the bi-directional case, the MAE and RMSE are lower with the model (with a gain of about 20%). The improvements of the MAE and RMSE with our model in both cases are statistically significant,

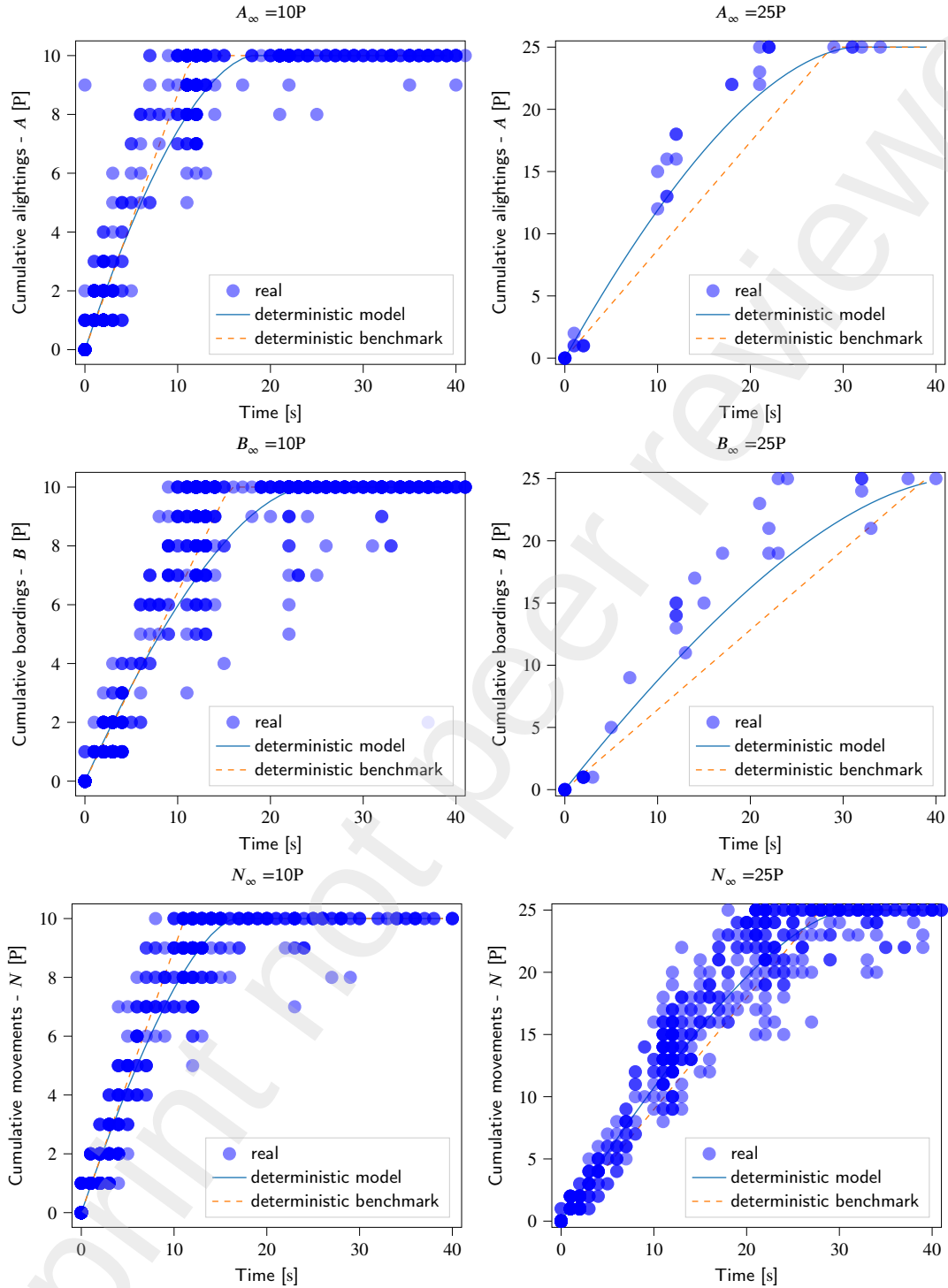


Figure 7: Examples of the profile of uni-directional cumulative alighting (top), uni-directional cumulative boarding (center), and bi-directional cumulative movements (bottom) from real data, the deterministic model, and the deterministic benchmark for processes having a total of 10 movements (left) and a total of 25 movements (right). The superiority of the model is visible for higher total volumes.

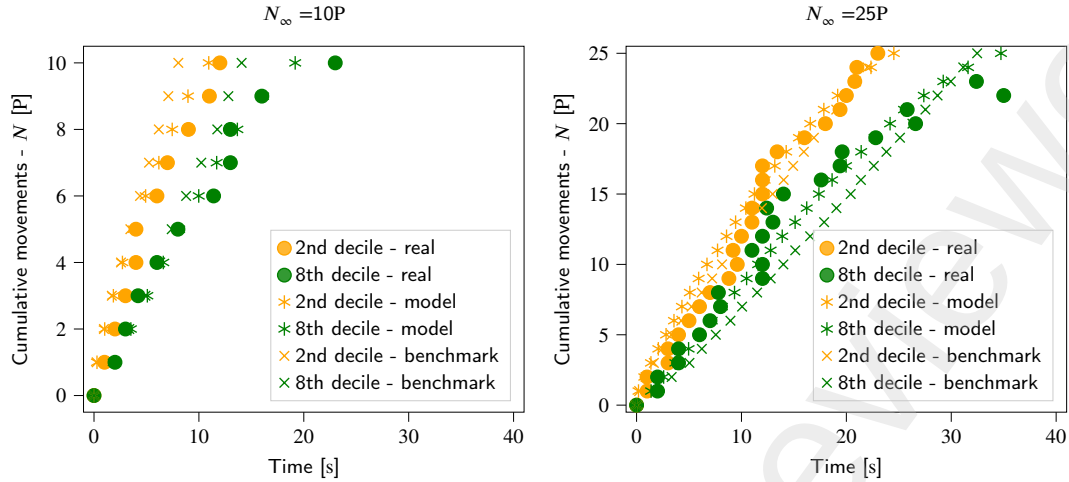


Figure 8: Examples of the 2nd and 8th deciles of times in the dynamic of bi-directional cumulative movements from real data, the stochastic model, and the stochastic benchmark for processes having a total of 10 movements (left) and a total of 25 movements (right). The superiority of the model is visible for higher total volumes and at the end of the alighting and boarding process.

Case	MAE [P · s ⁻¹]	RMSE [P · s ⁻¹]	Cost function - C_X [P ³ · s ⁻²]
Only alighting - Model	0.08 ± 0.002	0.16 ± 0.005	5.54 ± 0.23
Only alighting - Benchmark	0.10 ± 0.004	0.22 ± 0.006	4.98 ± 0.10
Only boarding - Model	0.11 ± 0.001	0.18 ± 0.002	2.61 ± 0.10
Only boarding - Benchmark	0.11 ± 0.002	0.21 ± 0.003	2.37 ± 0.04
Bi-directional - Model	0.10 ± 0.001	0.20 ± 0.002	6.92 ± 0.02
Bi-directional - Benchmark	0.12 ± 0.001	0.24 ± 0.002	6.15 ± 0.01

Table 5

Statistical performances of the fits of the deterministic model compared to the deterministic benchmark. The model does better in MAE except for uni-directional boarding and better in RMSE for all cases. The cost function is higher with the model because the cost function penalizes more at the beginning of the alighting process.

given the confidence intervals computed using the 5-fold cross-validation. For the uni-directional boarding case, no difference in MAE is observed. This result is consistent with the observations of Kuipers et al. (2025), who showed no influence of volume on boarding flow rates, contrary to alighting flow rates.

However, for all cases, the RMSE is much better for the model. This shows that larger errors are lower with the model. As the error is assessed on the derivative of A , B , and N , lower cumulative errors will be observed with the model, advocating for its superiority compared to the benchmark. Notice that the RMSE for the bi-directional case is higher than in both uni-directional cases; this can be due to hard-to-predict hindering of both directions of the flows that we have chosen to neglect.

In any case, the low width of the confidence interval computed thanks to the 5-fold cross-validation shows a low dependency of the fit quality on the data. In particular, the variance in the data quality does not influence the overall goodness of the fit.

The cost function is lower for the benchmark, which means that the beginnings of alighting and boarding processes at high volumes are better reproduced by the benchmark, which confirms the observations made in the previous section. This challenges one of the main hypotheses of our model, which is that the alighting and boarding process is only driven by macroscopic density effects. Yet, at the beginning of the process, a discontinuity is generated by the door opening. This is especially the case for only alighting or only boarding, where no one is standing on the other side of the door. The first alighting or boarding movement is not hindered by other passengers, resulting in faster flows. In addition, push-from-the-back effects should further increase the flow at the beginning in such cases.

The next section is dedicated to the assessment of the obtained alighting and boarding time distributions.

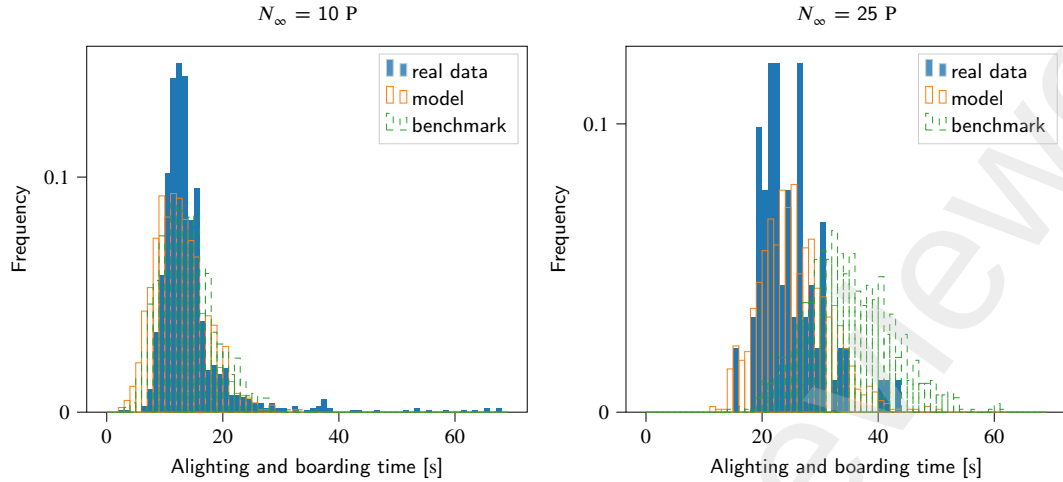


Figure 9: Distributions of alighting and boarding times for a total number of movements of 10P and 25P from real data, the stochastic model, and the stochastic benchmark. The stochastic model decently fits the distribution of alighting and boarding times from real data in both cases, while the stochastic benchmark only does so for 10 movements. Several outliers are observed in real data.

4.3. Alighting and boarding time distributions from the stochastic model

In this section, we compare the distribution of alighting and boarding times from real data to the simulated distribution obtained from the stochastic approach with both the benchmark and the model. Each distribution is computed for a fixed total number of movements, and the boarding/alighting ratio is neglected. We discuss only cases with a minimum of 30 values in the distribution. Hence, we focus on the bi-directional case, which is much more frequent in the dataset.

As a first illustration, Figure 9 shows the distributions of alighting and boarding times for a total number of movements of 10P and 25P from real data, the stochastic model, and the stochastic benchmark. For a total of 10 movements, the stochastic model and the stochastic benchmark have a similar distribution, both close to real data. For a total of 25 movements, the stochastic model remains close to real data, while the stochastic benchmark has a significantly different distribution. In both cases, a non-negligible number of outliers is observed in real data, whereas this is neither the case for the stochastic model nor for the stochastic benchmark. We have no insight into whether these outliers illustrate real-world situations or come from data issues, but the goal of our approach is to have a decent prediction for the central part of the distribution, representing the majority of real-world situations. Therefore, these outliers fall out of the scope of this analysis. This is why we focus on the 2nd and 8th deciles of the alighting and boarding times.

Figure 10 shows the evolution of the 2nd and 8th deciles of the alighting and boarding times distributions from real data and from the model. From this decile analysis, the model correctly fits the real data, especially for higher volumes. In particular, the interval between both deciles increases with volume, which is correctly captured by the model. The variability increases more slowly than the total number of movements, which is consistent with observations made by Buchmueller et al. (2008).

Figure 11 shows the Kullback-Leibler divergence in alighting and boarding times distributions between the model (respectively, the benchmark) and real data. The Kullback-Leibler divergence is lower with the model above 23 movements overall and below 12 movements overall, while being equivalent to the benchmark elsewhere. The lowest Kullback-Leibler divergences between the benchmark and real data are obtained for intermediate values (between 12 and 19 passengers). This is probably due to the cost function's form, which penalizes processes with few people overall (from the weighting factor), and processes with few observations to fit the data. Concerning the model, the Kullback-Leibler divergence decreases with the total number of movements. This highlights that the model's predictive power is slightly better at higher volumes, consistent with observations made in Figure 8.

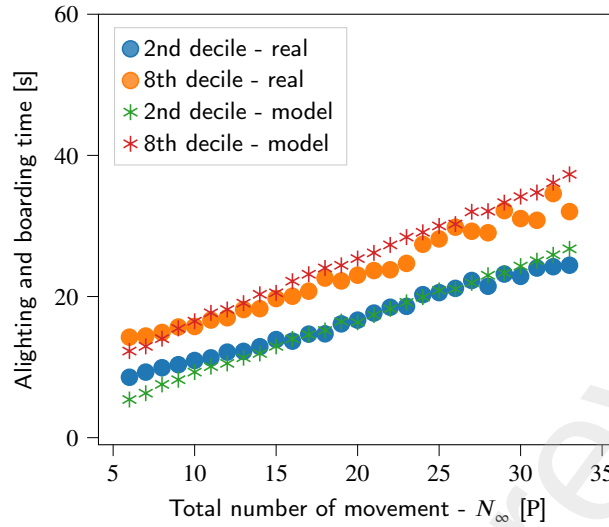


Figure 10: Evolution of the 2nd and 8th deciles in alighting and boarding times from real data (asterisks) and from the model (dots). Deciles from real data are well predicted by deciles of the model.

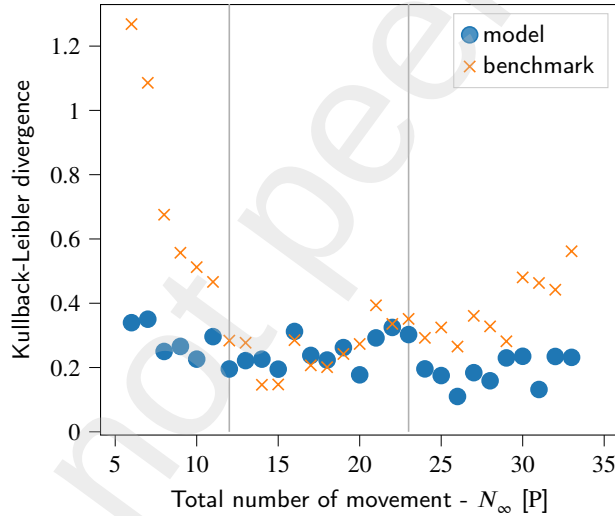


Figure 11: Kullback-Leibler divergence for the alighting and boarding times distributions between the model (in blue dots) or the benchmark (in orange crosses) and real data. The distributions with the model are closer to real data with a higher total number of passengers.

4.4. Physical interpretation of the parameters

The parameters obtained when fitting the model have physical meanings. As a reminder, $\psi = v_0 D \rho_{max}$, where v_0 is the free-flow speed, ρ_{max} is the maximal density, D is the door width (fixed). ρ_{max} could vary according to passenger body shapes (body size, body weight, luggage, etc). v_0 depends on people's behavior. With fixed γ , a higher value of ψ means a higher flow for a similar number of alighting, boarding, or movements; and, in particular, a higher maximal flow $\left(\frac{dX}{dt}\right)_{max}$. From the value of γ , it is possible to derive the critical demand X_{crit} (number of alighting, boarding, or movement corresponding to the highest flow). The critical demand can be obtained by deriving the left-hand side of equation (5) with respect to the variable $X_\infty - X$.

Case	$\psi = Dv_0\rho_{max}$ [P · s ⁻¹]	γ	$\left(\frac{dX}{dt}\right)_{max}$ [P · s ⁻¹]	X_{crit} [P]
Only alighting	9.1 ± 0.66	0.04 ± 0.003	2.28	300
Only boarding	5.0 ± 0.04	0.06 ± 0.001	1.25	133
Bi-directional	4.9 ± 0.04	0.09 ± 0.001	1.23	56

Table 6

Parameters obtained from the three fits done and their confidence intervals at 5% threshold. Parameters from the bi-directional and boarding flow are similar with similar confidence intervals. ψ is much higher for the alighting flow with a much larger confidence interval.

4.4.1. All data

Table 6 presents the parameters obtained with their confidence intervals. Values for the bi-directional flow are similar to those of the uni-directional boarding flow with the same confidence intervals. The parameter ψ for the alighting flow is higher with wider confidence intervals. This shows that the shape of the flow is more variable when there is only alighting. This is probably due to variability in the configuration of passengers inside the train before alighting (people in front of the door versus people seated further in the carriage), which should influence the alighting rate for a fixed number of passengers.

Though a higher value for the uni-directional alighting case, the width of the confidence interval remains low compared to the parameter. This shows that the variance in the data quality does not influence the fitted parameters.

For similar volumes, the uni-directional alighting rate is significantly higher than the uni-directional boarding rate. This result is consistent with the literature. For example, Fu et al. (2023) also found significantly higher flows for only alighting than for only boarding (in the case of a bi-directional flow with full compliance to the alighting priority). However, they found alighting rates closer to boarding rates in generic bi-directional flows as shown by Kuipers et al. (2025) or Seriani et al. (2019). This result is not reproducible with our approach as we amalgamate alighting and boarding in generic bi-directional flows. Similarly to how we amalgamate alighting and boarding in bi-directional flows, Fu et al. (2023) found movement rates closer to uni-directional boarding flows compared to uni-directional alighting flows, which is consistent with our results.

The critical demand that we obtain should be considered with caution. Figure 4 showed that no situation having volumes higher than the critical demand was observed in the data. This means that these critical demands are projections from low-volume situations. This also means that, based on our results, no congested regime is observed in the data. The critical demand is particularly high for uni-directional flows (alighting or boarding). Knowing the capacity of one car of the train is 132P, the model predicts that no congested regime may occur in any only alighting or only boarding situations. In the case of a bi-directional flow, it should be possible to see a congested regime, but data of higher demand would be necessary for further discussion.

In Table 7, the values of the obtained parameters ψ are compared with those obtained from comparable flow-density relationships in the pedestrian literature (i.e. Greenshields formulations). For only alighting, ψ is close to the values of level walkways in the literature, where ψ is generally higher. For only boarding or bi-directional flows, it is close to values for ascending stairs. The difference between uni-directional and bi-directional cases in the literature is generally low, and there is no trend on which type of flow should have a higher ψ than the other. This is consistent with the similarity observed in only boarding flows and bi-directional flows from our model. In further detail, uni-directional and bi-directional flows have different free-flow speed and maximal density parameters in the literature; these parameters cannot be obtained with our model.

Existing studies show a higher maximal flow in the fundamental diagram for pedestrian crossing bottlenecks compared to other walkways' results (Hoogendoorn and Daamen, 2005; Seyfried et al., 2009). This would lead to a higher ψ , which is not observed here. The bottlenecks studied in the work of Seyfried et al. (2009) were narrower than the 1.6m door studied here, which might be a potential explanation.

4.4.2. Specific sub-samples

For bi-directional flows (see Table 8), we compare the obtained parameters varying non-compliance, day-period, and for a specific station. A first observation is that the values are close for both parameters in all cases.

Compared to peak hours, ψ in off-peak hours is slightly lower, probably meaning a lower v_0 . Bosina and Weidmann (2017) showed a slight but not significant increase in walking speed in peak hours, in line with our result. γ is similar in

Case	Context	v_0 [m · s ⁻¹]	ρ_{max} [P · m ⁻²]	$\psi = Dv_0\rho_{max}$ [P · s ⁻¹]
Only alighting (our model)	Train door			9.1 ± 0.66
Only boarding (our model)	Train door			5.0 ± 0.04
Bi-directional (our model)	Train door			4.9 ± 0.04
Fruin (1971)	Level uni-directional	1.43	4.09	9.20
Fruin (1971)	Level bi-directional	1.36	4	8.70
Fruin (1971)	Ascending stairs	0.57	7.40	6.75
Fruin (1971)	Descending stairs	0.65	6.70	6.97
Nelson (2002)	Level	1.4	3.8	8.51
Nelson (2002)	Any stairs	0.8	3.8	4.86
Ye et al. (2008)	Level uni-directional	1.44	3.2	7.38
Ye et al. (2008)	Level bi-directional	1.19	4.11	7.83
Ye et al. (2008)	Descending stairs	0.97	6.06	9.41
Ye et al. (2008)	Ascending stairs	0.69	4.93	5.44

Table 7

Comparison of the parameters ψ from our model and comparable existing works (with Greenshields formulation). The parameter obtained for uni-directional alighting flow is close to the case of level walkways. Cases of uni-directional boarding or bi-directional flows are more in line with cases of ascending stairs.

	ψ_N [P · s ⁻¹]	γ_N [P ^{-1/2}]
All	4.9 ± 0.04	0.09 ± 0.001
Peak hours	4.9 ± 0.02	0.09 ± 0.0004
Off-peak hours	4.8 ± 0.05	0.09 ± 0.002
Lower non-compliance	5.0 ± 0.03	0.09 ± 0.001
Higher non-compliance	5.1 ± 0.16	0.07 ± 0.003
<i>Versailles-Chantiers</i>	4.8 ± 0.04	0.09 ± 0.001

Table 8

Comparison of the parameters for bi-directional flows between peak hours and off-peak hours; lower non-compliance and higher non-compliance; at the station of *Versailles-Chantiers*. $\psi = v_0 D \rho_{max}$.

both cases, which means no influence of peak hours on the relationship between volumes and density, and an increase in flow for similar volumes. Other works, such as the study of Wiggendaad (2001) showed a very low increase of alighting and boarding rates in peak hours, similar to here.

Regarding non-compliance, higher non-compliance has a not significantly higher ψ while having a lower γ . A lower γ means that the density increases more slowly with the remaining number of passengers to alight or board. This contradicts findings from Baali et al., where higher densities were found with higher non-compliance and the same volumes. The confidence interval of both parameters at higher non-compliance is wider than in any other case. This can be due to a more variable flow at higher non-compliance between one situation and another. For example, Yang et al. (2020) showed more fluctuation in the alighting and boarding dynamics for non-compliance cases from simulation results.

For *Versailles-Chantiers*, the parameters are very similar to the case with all the data. This confirms that stations have negligible to no influence on the modelling of the alighting and boarding process. *Versailles-Chantiers* being, moreover, the station with the highest demand in *Line N*. This confirms that the model considers intrinsically the volume and parameters are independent of it, which was a hypothesis of the approach.

5. Conclusion

In this work, we presented an innovative approach to model the alighting and boarding process. This approach differed from what is generally done by discussing the alighting and boarding process as a cumulative flow problem rather than a two-dimensional pedestrian movement. Two models were discussed: a deterministic model depending on alighting and boarding volumes fitted on disaggregated passenger counting data; and a stochastic model, which is

a Markov process whose average is the deterministic model. The disaggregated passenger counting data used were particularly suitable to calibrate the model. The deterministic model was then compared with a linear benchmark and showed better predictive power for uni-directional alighting flows and bi-directional flows (with a gain of around 20% in MAE and RMSE). The stochastic model helped capture the variability of the flow, and the probability distribution of alighting and boarding times for a given volume is correctly reproduced. The physical parameters obtained from the fit were in line with existing literature. In summary, the proposed approach is a data-driven stochastic model that precisely fits time-dependent alighting and boarding rates, and its parameters are physically consistent.

This novel approach is a relevant addition to existing models for the alighting and boarding process, provided disaggregated passenger counting data is available. One can suggest dwell time planning as an implication. Forecasting alighting and boarding time distributions at the door level would help assess train dwell times. This would enable building more robust schedules to reduce dwell time delays. Another implication concerns modeling injection flows in stations and trains for pedestrian simulations.

This model is built to deal with cumulative flows. Then, it is complicated to compare its results with two-dimensional pedestrian models from the literature, especially because no two-dimensional data (video) is available to calibrate two-dimensional models. Our model would benefit from a comparison to more sophisticated existing approaches, such as cellular automata or social force models. In addition, the data used for this work gave information on alightings and boardings every 3s on average. Such gaps between two counting events make the alighting/boarding rates rather imprecise, and obtaining a good fit is made more difficult.

In future work, we will extend the approach to other lines with higher demands, which will be possible with the recent arrival of new rolling stocks on the Paris RER. In this context, it will be crucial to consider on-board and on-platform people who neither alight nor board. Improvements of the model will be necessary, including a better modeling of bi-directional flows, with the addition of stochastic choices of whether the following movement is an alighting or a boarding. This addition will help consider different rates for alighting and boarding in bi-directional flows, and a direct consideration of boarding/alighting ratio that has been neglected so far, and whose influence will be significant for higher volumes. Other functions for the fundamental diagram and the density function will also have to be assessed to deal with higher congestion levels and to challenge the results presented here. The behavior at the beginning of the alighting and boarding process regarding discontinuity after the door opening will also be studied for better modeling. Finally, this approach will be compared to two-dimensional alternatives.

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Appendix: computation of the critical demand and the maximal flow

Critical demand The critical demand is the number of remaining movements (alighting and/or boarding) $X_\infty - X$ for which the flow is maximal. To compute it, we use the left-hand side of the equation:

$$\frac{dX}{dt} = \psi \left(1 - \exp(-\gamma \sqrt{X_\infty - X(t)}) \right) \left(\exp(-\gamma \sqrt{X_\infty - X(t)}) \right).$$

674 We compute the derivative of:

$$g(y) = \psi (1 - \exp(-\gamma \sqrt{y})) (\exp(-\gamma \sqrt{y})) ,$$

675 which is:

$$\frac{dg}{dy} = \frac{\psi \gamma}{2\sqrt{y}} (2 \exp(-2\gamma \sqrt{y}) - \exp(-\gamma \sqrt{y})) .$$

676 This derivative is zero for:

$$y = X_{crit} = \left(\frac{\log(2)}{\gamma} \right)^2 ,$$

677 which gives the critical demand.

678 *Maximal flow* The maximal flow is the flow at the critical demand, but it can be directly obtained from equation (5).

679 Writing $\mu_X = \exp(-\gamma \sqrt{X_\infty - X(t)})$, we have:

$$\frac{dX}{dt} = \psi (1 - \mu_X) \mu_X .$$

680 From there, we can get the maximal flow:

$$\left(\frac{dX}{dt} \right)_{max} = \psi \times 0.25 .$$